

Lecture 7: Introduction to Computer Graphics for AR/VR

“Why does VR feel real but video games feel less immersive?”

“What happens if VR lags by 1 second?”

Explain briefly:

- AR/VR is not just visuals → it's **real-time computer graphics**
 - Everything they see inside headset = generated by graphics pipeline
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What is Computer Graphics in AR/VR?

Explain concepts:

- Definition:

Computer graphics is the creation, manipulation, and rendering of visual content using computers.

In AR/VR context:

- Rendering 3D objects
- Simulating lighting and shadows
- Depth perception
- Real-time interaction

Examples:

- VR game → objects react instantly
 - AR filter → face tracking + real-time rendering
 - Industrial VR simulation
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Graphics Pipeline (Core Technical Part)

Teach with a simple diagram on board / slide:

Graphics Pipeline Flow:

1. **Application Stage (CPU)**

- o Game logic
- o Physics
- o Input handling (controller, headset tracking)
- 2. **Geometry Processing (GPU)**
 - o Vertex processing
 - o Transformations (model → world → camera → screen)
- 3. **Rasterization**
 - o Convert 3D objects into pixels
- 4. **Fragment / Pixel Shading**
 - o Colors
 - o Textures
 - o Lighting effects
- 5. **Frame Buffer → Display**
 - o Final image sent to headset screen

Simple analogy:

Like making a movie frame every 1/90th second.

Important keywords to emphasize:

- Vertex
- Shader
- Texture
- Frame buffer
- GPU

Role of Real-Time Graphics in AR/VR

Explain why AR/VR is different from normal graphics:

Normal Graphics AR/VR Graphics

Can tolerate delay Needs ultra-low latency

30 FPS okay Needs 60–120 FPS

Less immersive Highly immersive

No motion tracking Head & hand tracking

Discuss key points:

- Real-time rendering = response within milliseconds
- Latency causes:
 - o Motion sickness
 - o Disorientation
 - o Poor immersion

Key terms to introduce:

- FPS (Frames Per Second)
 - Latency (Motion-to-photon delay)
 - Performance vs realism tradeoff
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Quick Recap + Questions**Summarize:**

- Computer graphics powers AR/VR realism
- Graphics pipeline explains how image is formed
- Real-time graphics is critical for immersion

Quick questions to ask class:

- What is the purpose of rasterization?
- Why is low latency important in VR?
- Which component mainly handles rendering: CPU or GPU?